

Lecture 5

Property Rights, the Coase Theorem, and the Environment

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Property Rights, the Coase Theorem, and the Environment

Pigovian Taxes and Rights

- A **Pigovian tax** makes polluters pay for the external damages they impose on society.
- Implicit assumption:
 - Society has a **right to clean environment** and to be compensated for damages.
 - Polluters **do not have unlimited rights** to emit.
- Intuitively appealing, but raises a broader question:
 - Who **owns the right** in cases of environmental harm?
 - The polluter, or society at large?

A Wetland Example: Competing Rights



- Farmer **Alpha** drains a wetland to grow crops → earns **\$5,000 profit**.
- Downstream neighbor **Beta** loses **\$8,000** in crop damage from flooding.
- **Question:**
 - Does Alpha's property ownership give him the **right to drain**?
 - Or does Beta (or society) have the **right to protection** from harm?

Negotiation with Alpha Holding Rights

- Suppose Alpha **has the legal right** to drain.
 - Beta can offer to pay him to keep the wetland intact.
 - Beta's **willingness to pay**: up to \$8,000 (her avoided damages).
 - Alpha's **willingness to accept**: at least \$5,000 (his foregone profit).
 - This creates a **bargaining range**: **\$5,000–\$8,000**.
- *Example outcome*: Beta pays Alpha **\$6,000**.
 - Alpha gains \$1,000 more than draining.
 - Beta avoids \$8,000 damage, but still pays \$6,000 → net \$2,000 better off.
 - The actual payment depends on each side's **bargaining power**.
- **Efficient result**: wetland preserved.

Negotiation with Alpha Holding Rights

Why Beta Could Still Be “Unhappy”

- Beta is **better off** than if Alpha drained (\$2,000 net gain).
- But she could still **feels burdened** because:
 - She pays a large sum to prevent harm.
 - She may believe she should not have to pay at all.
- Lesson: **Efficiency $\neq$ Equity**.
 - Efficient outcomes can still feel unfair.

Negotiation with Beta Holding Rights

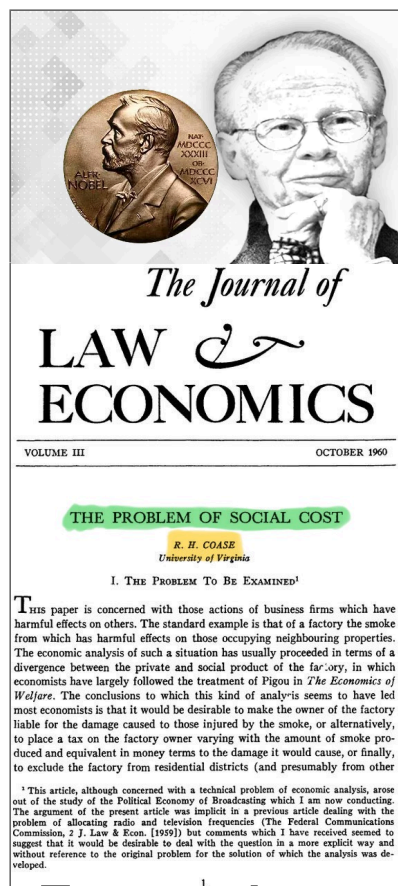
- Suppose the law gives Beta the **right to a wetland**.
 - Alpha must buy permission to drain.
 - With current values:
 - Alpha can offer up to \$5,000 (his potential profit).
 - Beta demands at least \$8,000 (her avoided loss).
 - No agreement possible → wetland preserved (**same efficient outcome**)
- Outcome is identical regardless of who holds the rights —Wetland stays intact.

Negotiation with Beta Holding Rights

A Change in Payoffs

- Imagine a *new crop* grows well on drained wetlands.
 - Alpha's profit rises to **\$12,000**.
 - Beta still faces \$8,000 in damages.
- Now negotiation is possible:
 - Alpha can pay Beta between \$8,000 and \$12,000.
 - *Example*: \$10,000 payment →
 - ▷ Alpha nets \$2,000.
 - ▷ Beta nets \$2,000.
- Takeaway: **Efficient outcome depends on relative benefits and costs**, not on initial assignment of rights.

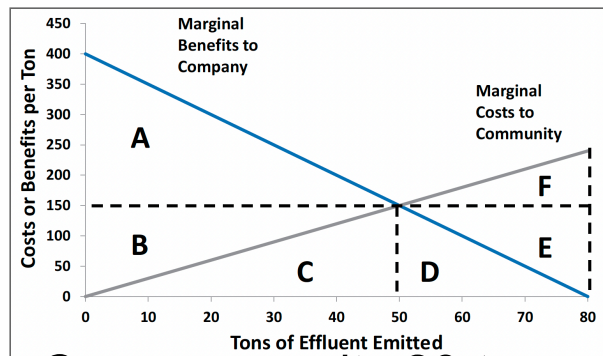
The Coase Theorem



- **Transaction costs:** Expenses of making a deal or negotiation (e.g., legal fees, administrative costs, or costs of bringing parties together).
- **The Coase Theorem:**
 - If property rights are **well defined** and **transaction costs are negligible**,
 - Then private bargaining will lead to **efficient allocation** of resources,

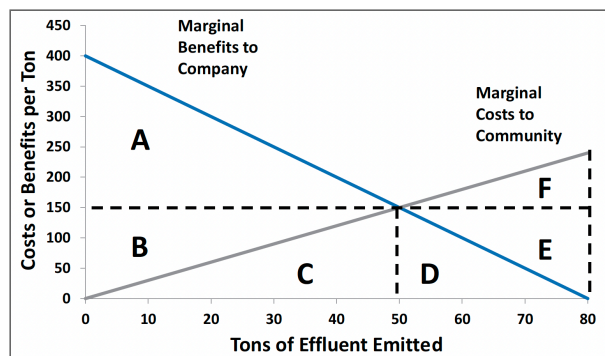
- Even when **externalities** are present.
- Rights determine **who pays whom**, but **not the final efficient outcome**.
- Assumes: Perfect information; Few parties; No high legal or enforcement costs.

Welfare Analysis of Company vs. Community



- Company emits **80 tons of effluent** into a river.
 - Company gains profits (MB of pollution).
 - Community bears damages (MC of pollution).
- Efficient solution: **50 tons**, where $MB = MC$.
 - Beyond 50, extra damage $>$ extra profit.
 - Below 50, extra profit $>$ extra damage.

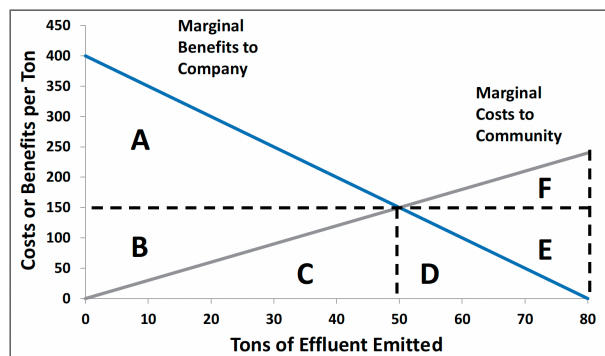
Negotiation with Community Holding Rights



- Assume:
 - Pollution rights sell for **\$150** each.
 - The **community** owns the rights.
- At 50 tons: $MB = MC = \$150$.
 - Company's payments = **\$7,500** (area B + C).
 - Company benefits = **\$13,750** (area A + B + C).
 - Total community damages = **\$3,750** (area C).
- Net results:

- Company' Net Benefit: **\$6,250.**
- Community's Net Benefit: **\$3,750.**
- **Social Welfare = \$10,000.**

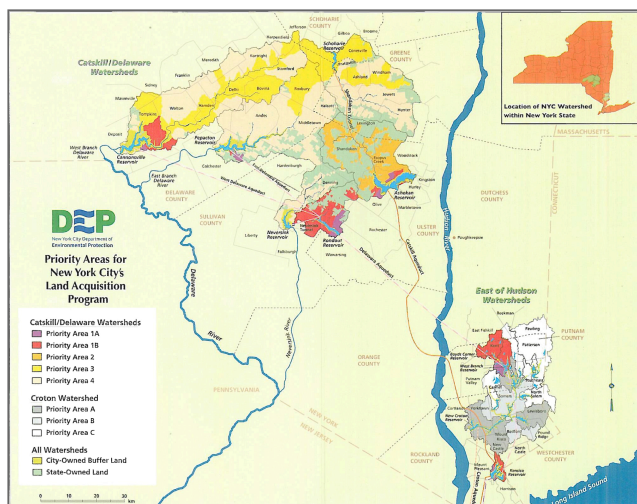
Negotiation with Company Holding Rights



- Assume:
 - Pollution rights sell for **\$150** each.
 - The **company** owns the rights.
- At 50 tons: $MB = MC = \$150$.
 - Community's payments = **\$4,500** (area D + E).
 - Company benefits = **\$13,750** (area A + B + C).
 - Total community damages = **\$3,750** (area C).
- Net results:

- Company's Net Benefit: **\$18,250.**
- Community' Net Benefit:
-\$8,250.
- **Social Welfare = \$10,000.**
- **Distribution of welfare differs.**

Case Study: NYC Land Acquisition Program



- NYC must provide clean drinking water for **8.4 million people**.
 - Option 1: Build massive water filtration plants (very costly).
 - Option 2: **Protect watersheds upstream** by buying land & easements.
- NYC purchased property rights *voluntarily* (no eminent domain used):
 - Acquire 355,000+ acres from willing sellers; Pay **fair market value**.

- Cheaper than filtration; Clean water maintained;
Example of **Coasean bargaining** on a large scale.

Property Rights and Takings

- **Eminent Domain:** Gov't may take land for public use but must pay compensation (Fifth Amendment).
- **Taking:** Any gov't action that deprives an owner of property rights.
 - **Regulatory Takings:** Laws that restrict land use and may reduce property value.
- **Supreme Court Ruling – *Lucas v. South Carolina Coastal Council*:**
 - South Carolina Law barred building on Lucas's beachfront lots.
 - *Ruling:* A regulation that removes **all economic use** of land is a **taking** → compensation required.

Type of Taking	Definition	Compensation?	Frequency
Total Taking	All economic use eliminated	 Yes	Rare
Partial Taking	Some economic use remains	 No	Common

- **Q:** What is the rationale for not requiring compensation in cases of **partial takings**?

Limits of the Coase Theorem

- **The Promise of Coase**

- In theory, if property rights are clearly assigned, parties can negotiate efficient outcomes without government intervention.
- This idea underpins **free market environmentalism**—the belief that expanding property rights and market mechanisms can handle pollution control.

- **The Reality: Transaction Costs**

- Efficient bargaining assumes low transaction costs, but in practice these costs can be very high.
- When only two parties negotiate (e.g., Alpha and Beta), costs are manageable.
- With 50 downstream communities and multiple factories, negotiation becomes cumbersome, maybe impossible.

Limits of the Coase Theorem

- **Free-Rider Effect**

- Some communities may wait for others to pay off the polluter, hoping to enjoy benefits without contributing.
- This leads to *undersupply of pollution reduction*.
- **Example:** if one community refuses to contribute, others may hesitate too, and no deal is reached.

- **Holdout Effect**

- When many parties must agree, a single holdout can block progress by demanding more compensation.
- **Example:** 49 communities agree, but the 50th demands excessive payment, threatening to veto the entire agreement.
- Like free-riding, this prevents efficient negotiation.

Limits of the Coase Theorem

- **Nonhuman and Ecosystem Impacts**

- Many environmental damages harm ecosystems, plants, or animals rather than identifiable individuals.
- Who bargains for the survival of birds, bees, or endangered species?
- Groups like *The Nature Conservancy* buy land to preserve ecosystems, but their reach is limited.
- In markets, ecological values often lose to economic ones.

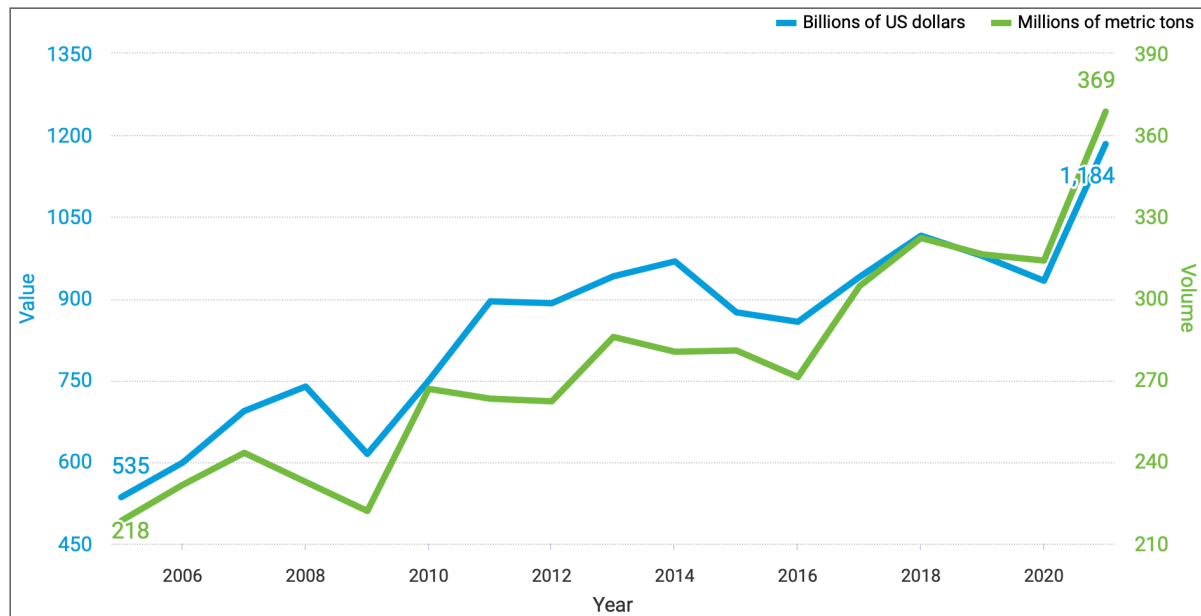
- **Future Generations**

- Property rights usually apply to the current generation.
- Climate change, fisheries depletion, and species extinction impose costs on future people with no voice in today's negotiations.
- Markets alone cannot ensure *intergenerational fairness*.

Limits of the Coase Theorem - Equity Concerns

- Poor communities may lack the resources to “buy out” polluters, even when health damages are severe.
- They might accept hazardous facilities out of desperation for compensation.
- Meanwhile, wealthier communities can afford to preserve open space, while poorer ones cannot.
- Example: **Global plastic waste trade**
 - High-income countries export waste to poorer nations that accept environmental risks for financial gain.
 - A large share of traded plastics is mismanaged.
 - For decades, China was the largest importer, receiving 72% of global plastic waste exports and mismanaging about 76% of it (EIA, 2021).

Global plastics trade hits record \$1.2 trillion



Global plastics trade hits record \$1.2 trillion



Environmental Justice

- Defined by EPA: *“Fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income.”*



Tap water in a Flint hospital on Oct. 16, 2015.
JOYCE 218 / FLINTWATERSTUDY.ORG

- **Example: Flint, Michigan Water Crisis (2014–)**
 - City switched to Flint River water, which was corrosive and not treated properly → lead contamination.
 - 6,000–12,000 children exposed.
 - Community largely low-income and African-American.
- Lesson: **Markets alone cannot ensure fairness** in environmental outcomes.

Environmental Justice

	Within 1 km	Between 1 km & 3 km	Between 3 km & 5 km	Beyond 5 km
Percent people of color	47.70%	46.10%	35.70%	22.20%
Percent poverty rate	20.10%	18.30%	16.90%	12.70%
Mean household income	\$31,102	\$33,118	\$36,920	\$38,745

Race, Poverty and Hazardous Waste Facilities

- Do polluting industries **choose to locate** in poor and minority communities—where land is cheaper and political resistance is weaker?
- Or do **low-income households move in** near toxic storage or disposal facility (TSDF) after they are built, attracted by lower housing costs?
- **Pastor et al. (2016), “Which Came First? Toxic Facilities, Minority Move-In, and Environmental Justice”:**
 - A **10% point increase** in minority population share was linked to **~34–39% higher odds** of a TSDF siting.
 - The **pace of increase** in minority residents near TSDFs was similar to the pace elsewhere in the region.